

1 Basics of the Group Bonus

Summary

The group-related incentive wages are used to comprise machines and/or workplaces to so called "premium groups".

By default, either a payment scheme depending on the utilization or depending on the performance efficiency rate can be assigned to a premium group. For the utilization bonus the efficiency of the machine usage time is decisive, and for the performance efficiency rate the produced quantity related to the standard time.

When the formula-based premium/ incentive wage is used, it is also possible to realize by customer-specific formulas also other calculation methods and premium modes.

Upon the login to a machine and/or workplace, the order processing data and employees will be assigned to a premium group for the processing duration.

The corresponding activities are assigned once per month to the persons on a pro-rated base to the presence times that the persons worked in the individual premium groups.

Calculation and availability of the data

The day data on premium groups will be determined and compressed through the wage calculation. These day data serve as basis for the monthly results of the premium groups. Modifications to the ADE postings will be transferred to the premium group results during the daily night run.

HYDRA will maintain these daily group results in the time domain that can be specified in the data management and automatically delete them once this period has elapsed.

The monthly results of the premium groups are calculated directly when the list of the current day data is called.

Calculation of the incentive bonus

The incentive bonus is based on the quantities that were produced in the group to one specific order. All order and personnel postings are used for the calculation of the incentive bonus that were collected at the machines of a premium group in a given settlement period (month).

The product of the quantity of interruption and end postings and of operation-related standard time (t_i and t_e) per piece is the total standard time. The ratio of this standard time to the really needed personal processing time of the group defines the group's performance.

The incentive wage determination is based on the following method:

$$\text{Performance level } LG = \frac{\sum_{Ops} t_r + m * t_e + \sum Bonuses_{Targettime}}{\sum_{Persons} Processing\ times - \sum Bonuses_{Actual\ time}}$$

- The performance level is determined per premium group.
- t_r is the set-up standard time for the operation that was stored to the HYDRA order backlog at the time of processing. If necessary, this value will be entered via the PPS interface and can then be overwritten in the corresponding field of the HYDRA order backlog. The value will be used from the order postings.
- t_e is the individual standard time for the operation that was stored to the HYDRA order backlog at the time of processing. If necessary, this value will be entered via the PPS interface and can then be overwritten in the corresponding field of the HYDRA order backlog. The value will be used from the order postings.
- m is the yield that was posted during the processing of the operation in HYDRA (if necessary through several persons). The value will be used from the order postings.
- Bonuses are created in HYDRA depending on the order and operation and on the premium group. All bonuses that are stored for the premium group in HYDRA will be accounted for in the calculation. Depending on the bonus configuration, bonuses on top of the fraction stroke will be applied as "bonus" on the target time and those below as "credit note" to the actual time.
- The processing times of everyone working at the workplaces of the premium group operations (only production orders and no overhead cost orders) in the calculation period will be comprised in the denominator. In general the pro-rated personnel durations will be compensated if one person processed several operations in parallel. The sum results from the personnel postings (B-records).

Calculation of the utilization bonus

The ratio of standard time to the total runtime of all processed orders/ operations as well as of the authorized overhead cost times defines the utilization degree (NG).

$$NG = \frac{\sum_{Ops} t_r + m * t_e + \sum Bonuses_{Targettime}}{\sum_{ProductionOps} Machine\ usage + \sum_{unauth.OG-OPs} unprod.times - \sum Bonuses_{Actual\ time}}$$

- The utilization degree is determined per premium group.

- t_r is the set-up standard time for the operation that was stored to the HYDRA order backlog at the time of processing. The value will be used from the order postings.
- t_e is the individual standard time for the operation that was stored to the HYDRA order backlog at the time of processing. The value will be used from the order postings.
- m is the quantity that was posted during the processing of the operation in HYDRA (if necessary through several persons). Depending on the setting in the *LLE setup* this quantity can be included in scrap or not. The value will be used from the order postings.
- The machine occupation times will be added up via all production orders. The sum results from the order postings.
- Unproductive times result from postings to a (permanent) overhead cost order of general validity (e.g. cleaning of machines, lack of orders), or from machine waiting period postings. All these times will **only then** be taken into account in balancing when they are posted to the resource performance account **U8**. For the machine waiting period there is no overhead cost (OC) order in HYDRA.
 - In the default HYDRA settings the **machine waiting period** can be configured thus that it will automatically be posted to the account U8 and thus be accounted for in the formula **without further action as unproductive time**.
 - For **overhead cost (OC) postings** the times can initially be posted to other RPAs so that they will **initially apply as authorized downtime** and will not show any effect in the above formula. The foreman may then **repost these times manually** to the RPA U8 to classify them as **unauthorized downtime**. Without such a reposting through the foreman the downtimes from overhead cost orders will always be deemed authorized.
- Bonuses are created in HYDRA depending on the order and operation and on the premium group. All bonuses that are stored for the premium group in HYDRA will be accounted for in the calculation. Depending on the bonus configuration, bonuses on top of the fraction stroke will be applied as "bonus" on the target time and those below as "credit note" to the actual time.
- For different downtime reasons, the foreman can issue different (permanent) overhead cost time tickets to the operator.
- For all machines belonging to a utilization premium group, only one order can be logged in to a machine. Several operations cannot be processed in parallel at these machines.

Postings table as overview

The following table shows which postings are integrated in the calculation formulas:

Posting	Utilization bonus	Incentive bonus
Production orders (order)	1) Calculation of the standard time from produced quantity of pieces and time rules ($m \cdot t_e + t_r$) in the numerator 2) Determination of the machine usage time (denominator)	Calculation of the standard time from produced quantity of pieces and time rules ($m \cdot t_e + t_r$) in the numerator
Production orders (person)	- Will not be processed in the incentive wage determination-	Processing times (denominator)
Overhead cost order (order)	<i>Times only to RPA U8 (e.g. by manual reposting):</i> Integrated as unproductive times in the machine usage time (denominator). <i>Times outside the RPA U8 (e.g. default usage):</i> Will not be integrated into the formula since they are perceived as authorized downtimes. These times will, however, be shown separately and also be transferred in the interface.	- Will not be processed in the incentive wage determination-
Overhead cost order (person)	- Will not be processed in the incentive wage determination-	The sum of the overhead cost durations will not be integrated into the calculation formula, but be shown in HYDRA and be transferred in the interface. Operation numbers for which the first digit is smaller than 5 will be interpreted as "premium average". Operation numbers for which the first digit starts with 5 will be interpreted as "waiting times from GK".
Waiting period posting (machine/ OP)	Will be processed as "overhead cost order (order)". The HYDRA basic settings can be used to ensure that these postings will by default be used on the RPA U8 and correspondingly be directly counted as unproductive times.	- Will not be processed in the incentive wage determination-
Waiting period posting (person)	- Will not be processed in the incentive wage determination-	- Will not be processed in the incentive wage determination-
Bonuses	<i>Depends on the configuration of the bonus reasons:</i> As <i>bonus</i> on the standard time in the numerator and/or as <i>credit note</i> to reduce the machine usage time in the denominator	<i>Depends on the configuration of the bonus reasons:</i> As <i>bonus</i> on the standard time in the numerator and/or as <i>credit note</i> to reduce the processing time in the denominator

Additional premium formulas

When the formula-based premium/ incentive wage is used, it is also possible to realize other customer-specific calculation methods and premium formula.

Both premium forms "incentive bonus" and "utilization bonus" will be executed as default calculation by HYDRA if the premium group is configured correspondingly and can be overwritten by customer-specific formulas.

Premium areas

Premium groups can be comprised to premium areas. In doing so a premium group can be assigned to several areas so that quasi hierarchical structures can be mapped.

Configuration example:

Area 001 = Group A, B, C and D

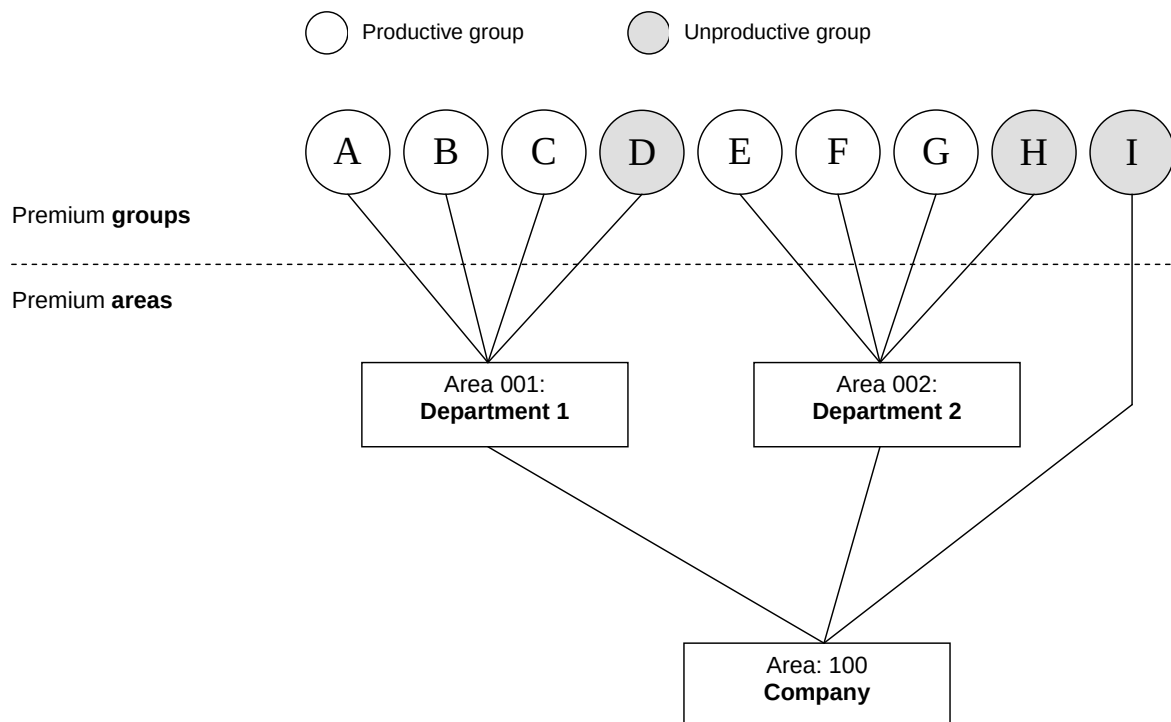
Department 1

Area 002 = Group E, F, G and H

Department 2

Area 100 = Group A, B, C, D, E, F, G, H and I company

This leads to a hierarchical structure:



Example:

The production is subdivided into several assembly lines that are paid as autonomous premium groups in group incentives. Conveyors transport the required materials to all production workplaces. Using the additional function "premium areas" it is possible to assign the conveyors to an "unproductive" group. This "unproductive" group and the assembly lines will be comprised to a premium area "production", in which the results of the individual groups will be collected and an average result be calculated. The "unproductive" conveyors will then be paid in accordance with the result from the premium area.

"Premium areas" are created as special types of premium groups in the master data. The master data of the premium group define also when a premium group must be paid according to the result of a premium area or of a different premium group.

Calculation of the area result

The result of a premium area is calculated as the result of a premium group through the wage calculation of a day. It is calculated from the day results of the assigned premium groups.

By default the individual factors such as *standard time*, *bonuses*, *unprod. time*, *duration*, *downtime*, *OC-time*, ... will be added up from the individual premium groups irrespective of the *type* of the premium group. The performance level of the premium area is calculated according to the formula

$$\text{Performance level} = \frac{\text{Standard time} + \text{bonuses}}{\text{Duration} - \text{credits}} * 100\%$$

If there is an additional function "formula-based bonus/ incentive wage" the calculation of the area result may vary depending on the customer's requests.